

Homework 1

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A.a. Find the names of all employees whose salary is higher than 500K.

$\pi \text{ Ename } (\sigma \text{ salary} > 500000 (E))$

b. Find the names of all employees work more than 15 hours on project x (Pname):

$\pi \text{ Ename } (\sigma \text{ Hours} > 15 (E \bowtie W \bowtie \sigma \text{ Pname} = "x" (P)))$

c. Find the ID's of those who work on a project located in Brooklyn but never in a project located in Lona Island.

$$\pi \text{ Essn } (W \bowtie (\pi \text{ Pnumber } (\sigma \text{ Plocation} = \text{"Brooklyn"}(P)) \\ + W \bowtie (\pi \text{ Pnumber } (\sigma \text{ Plocation} = \text{"Long island"}(P))))$$

d. Return the names of those who have the lowest salary among all employees who age is between 35 and 40.

$$\pi \text{ Ename } (\sigma 35 < \text{Age} < 40 (E)) \div \min (\sigma 35 < \text{Age} < 40 (E. \text{salary}))$$

e. Find the names of employees whose salary is higher than that of his/her supervisor

$$\pi E1. \text{Ename } (E1) - (E1 \bowtie E2. \text{SSn} = E1. \text{supervisor_} \\ \text{ssn} \wedge E1. \text{salary} \leq E2. \text{Salary } (E1, E2))$$

f. Find the names of all employees who have a dependent with the same name as themselves.

$$\pi E. Ename (E) \cap (E \bowtie E. Ssn = DE. Essn \wedge E. Ename = DE. D_name (E.DE))$$

g. Find the names and SSN's of employees who worked on every project located in brooklyn.

$$\pi E. Ename, E.Ssn (E) \cap (E \bowtie W. Essn = E.Ssn = E.Ssn) \bowtie (P. Pnumber = W.pno \wedge P.Plocation = "Brooklyn") (E, W, P)$$

h. Find the names of those who

worked on exactly 3 projects

$$\pi_{ESSN, Pno} (4) \rightarrow T_1$$

$$ESSN \Join \text{count_pno} (T_1) \rightarrow T_2$$

$$\sigma_{\text{count_pno} = 3} (T_2) \rightarrow T_3$$

$$\pi_{ESSN} (T_3) \rightarrow T_4$$

$$T_4 \Join E \rightarrow T_5$$

$$= \pi_{Ename} (T_5)$$

B. i. $\pi_{P,R}(\sigma_{R>S}(r))$

P	R
15	8
23	6

j. $r \times s$

r.S	r.B	r.R	s.A	s.B	s.C
10	a	5	10	b	b
10	a	5	25	c	3
10	a	5	10	b	5
15	b	8	10	b	6
15	b	8	25	c	3
15	b	8	10	b	5

25	a	6	10	b	6
25	a	6	25	c	3
25	a	6	10	b	5

k. $r \bowtie s$

r.P	r.B	r.R	s.A	s.C
15	b	8	10	6
15	b	8	10	5

l. $r \bowtie r.R < s.C^s$

r.P	r.B	r.R	s.A	s.B	s.C
10	a	5	10	b	b

m. $r \bowtie r.p = SA \wedge s.c \leq r.R^s$

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r.p	r.B	r.R	s.A	s.B	s.C
10	a	5	10	b	5
25	a	b	25	c	3

$$n. \pi_{B,R}(r) \div (\pi_R \sigma_{R < 8}(r))$$

↓

B	R
a	5
b	8
a	b

$\div$

R
5
6

$$o. r \div s$$

$\emptyset$

$$p. r - s$$

1 1 -

r.P	r.B	r.R
10	a	5
15	b	8
25	a	6

$$q. \pi_{P,R}(r) \cap \pi_{A,C}(s)$$

r.P	r.R
10	5